

WASP-80b

R-2078

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-80_180713 · created 2026-08-09T23:53:03+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2458312.7964 +/- 0.0028 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.069 +/- 0.066
Transit depth (Rp/Rs)^2	0.47 +/- 0.91 [%]
Orbital Inclination (inc)	88.3 +/- 1.72
Airmass coefficient 1 (a1)	0.75 +/- 0.16
Airmass coefficient 2 (a2)	0.17 +/- 0.17
Scatter in the residuals of the lightcurve fit is	21.45 %
Best Comparison Star	#1 - [538, 304]
Optimal Method	PSF photometry
Transit Duration (day)	0.04 +/- 0.01

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.069 ± 0.066 vs 0.1714 (deviation 1.25σ)
Duration fit vs published	0.04 d vs 0.0892 d (deviation 3.96σ)
Mid-time O-C	-0.9 min
Depth SNR	0.8
Residual scatter	21.45 %

Light curve

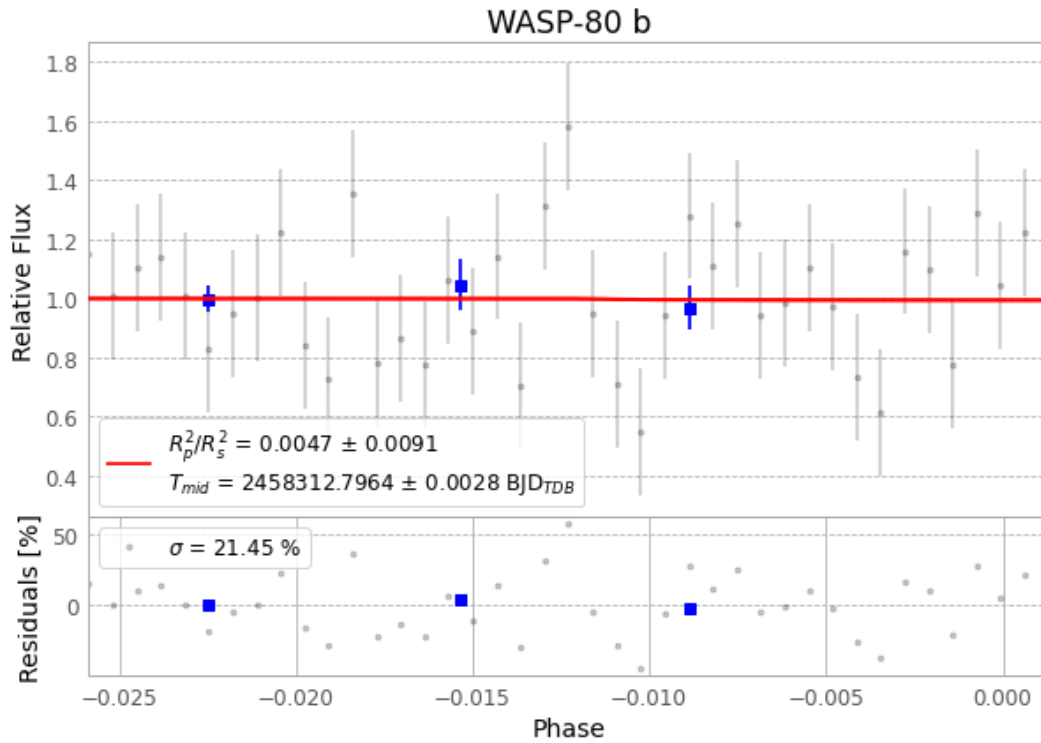


Figure 1 — FinalLightCurve_WASP-80 b_2018-07-12.png (SHA-256 cc17617f1ecb12f0...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [400, 127], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 a8337be432c270e0...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit fd74a91

Data provenance

41 FITS frames (27 MB), WASP-80180713050312.FITS → WASP-80180713070312.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-80-180713.log (09254d58181a...), FinalLightCurve_WASP-80 b_2018-07-12.png (cc17617f1ecb...), FinalLightCurve_WASP-80 b_2018-07-12.csv (2f9cab94d26...)

Compute resources

Wall time 131.7 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-09 23:53Z.